

From 30K sessions to better decisions

A portfolio case study covering data preparation, SQL analysis, modeling, Power BI dashboards, and evidence-led recommendations.

5,000 customers · 300 products · 4,558 orders · 486 returns



Where is the customer journey losing value?

The project turns a broad e-commerce question into measurable funnel, customer, product, delivery, and return questions.

01 Business problem — Identify where customers drop out, which segments drive revenue, and what the team should fix first.

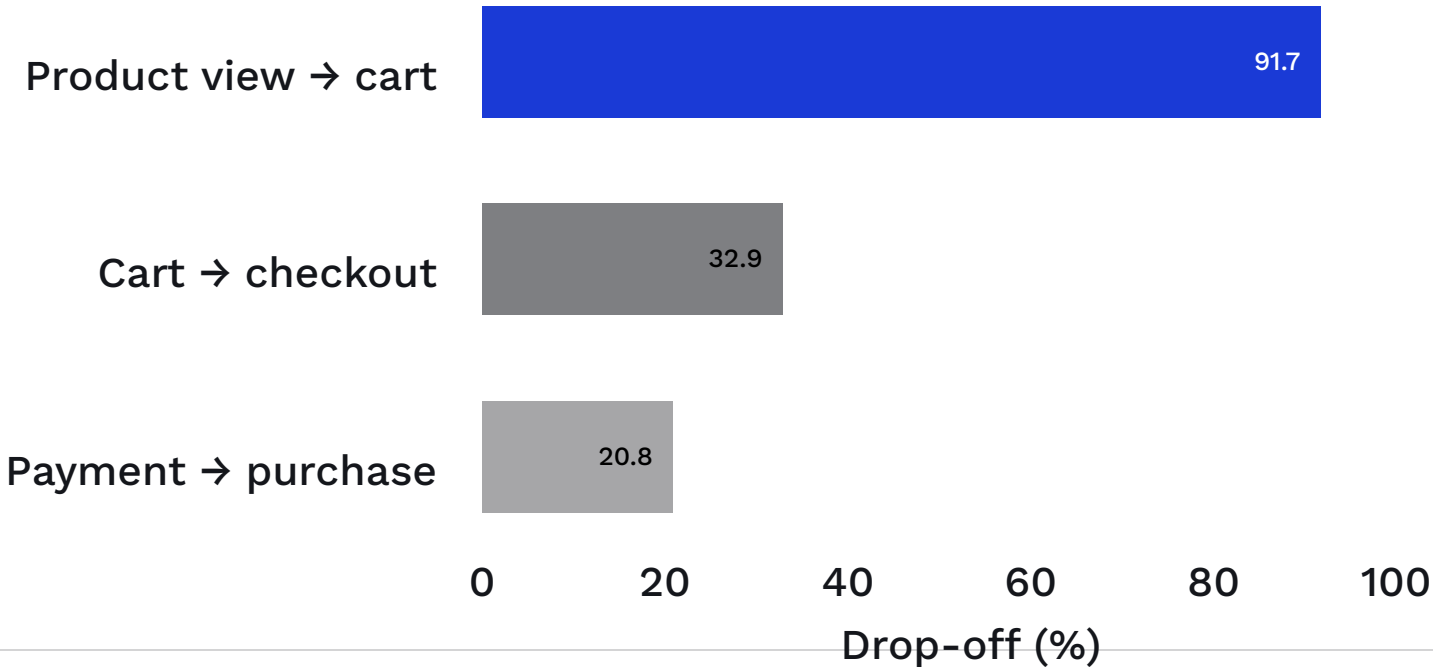
02 Source data — 5,000 customers, 300 products, 30,000 sessions, 4,558 orders, 4,558 delivery records, and 486 returns.

03 Analysis frame — Measure conversion, abandonment, revenue mix, returns, and delivery timing in one connected project.

The funnel exposed a 91.74% view-to-cart drop-off

DROP-OFF BY STAGE

Stage



15.19%

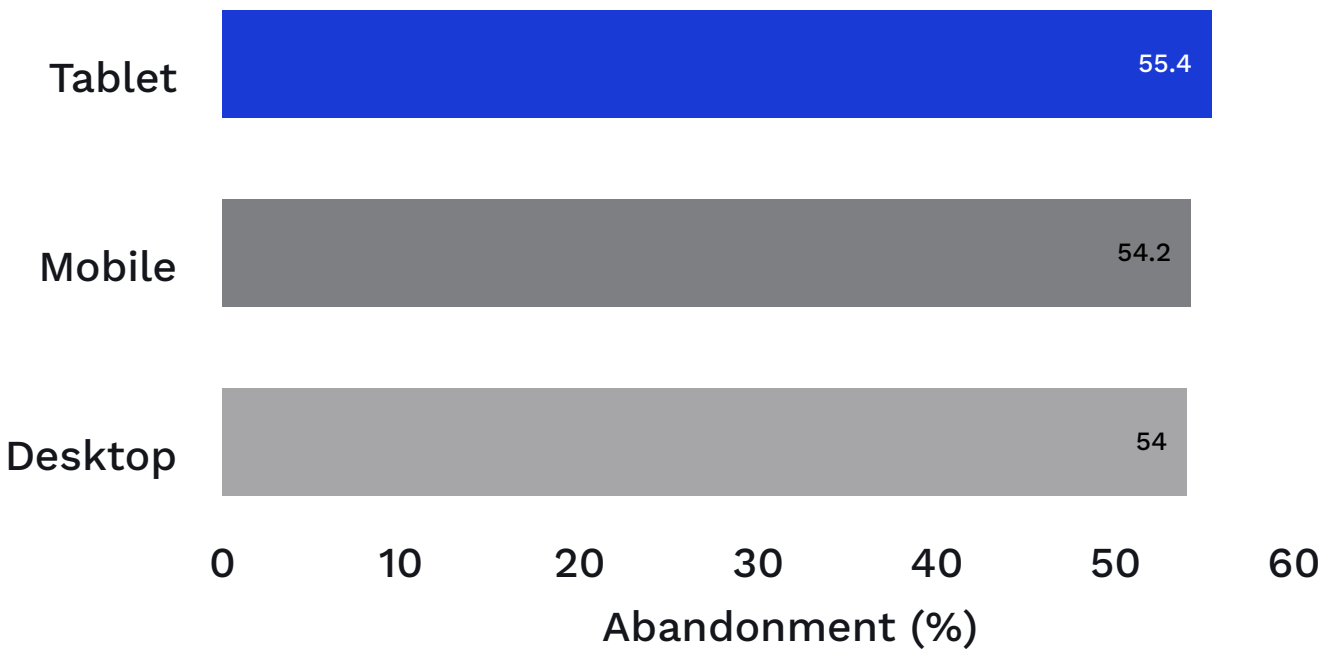
Overall conversion from session to purchase.

Only 8.26% of product viewers progressed to cart; later friction remained at cart-to-checkout and payment-to-purchase.

PRIORITY: IMPROVE THE PRODUCT-TO-CART EXPERIENCE, THEN TEST CHECKOUT AND PAYMENT FLOW

Abandonment was high across devices, categories, and sources

Device



55.41% — Tablet, the highest device rate

56.87% — Home & Kitchen, the highest category rate

55.81% — Instagram, the highest traffic-source rate

OVERALL: 54.23% ACROSS 9,958 CART SESSIONS AND 30K TOTAL SESSIONS.

Revenue was concentrated in returning customers and Electronics



Commercial picture: 83.15M revenue from 4,558 orders; AOV was 18.24K.

Customer value: returning customers generated 53.7M versus 29.5M from new customers.

Category and channel: Electronics generated 51.3M; Direct had the highest conversion at 15.94%.

Quality and delivery: Fashion had an 11.81% return rate, while conversion fell to 11.73% at 8 estimated delivery days.

I built the analysis from raw tables to executive views

01

Python + pandas

Loaded and profiled the tables, cleaned values, checked missing data, engineered age groups, converted dates, and loaded the prepared data into the database.

02

SQL Server

Answered funnel, customer, product, delivery, return, and conversion questions through joins, aggregations, and validation queries.

03

Fact + dimensions

Created a reusable model that connected event and transaction facts to customer, product, date, delivery, and return dimensions.

04

Power BI

Built an executive overview, funnel/drop-off view, and cart-abandonment dashboard to make the findings usable for decisions.

The recommendations were tied to measurable friction

Lift view-to-cart

Review product images, reviews, pricing, stock, and quantity information against the 91.74% product-view-to-cart drop-off.

Fix tablet checkout

Audit tablet UX and payment friction first: tablet abandonment was 55.41%, above mobile and desktop.

Retain valuable customers

Test retention journeys and first-order incentives, using the 53.7M returning-customer revenue as the value signal.

Improve traffic quality

Investigate Instagram landing experiences and channel intent; Instagram abandonment was 55.81% while Direct converted at 15.94%.

Reduce operational leakage

Investigate Home & Kitchen abandonment, Fashion returns, and delivery-time effects before scaling demand.

A complete analytics case study

The value is not just the dashboard; it is the path from messy data to a decision-ready story.

- Prepared and validated e-commerce data with Python and pandas.
- Used SQL Server and a fact/dimension model to make funnel, revenue, customer, product, delivery, and return metrics consistent.
- Translated the evidence into Power BI views and recommendations tied to measurable tests.

What the project shows: 15.19% overall conversion, 54.23% cart abandonment, 83.15M revenue, and clear priorities for product, checkout, retention, marketing, and operations.

Evidence base: e-commerce project pages 1–4, business problem document, and cart-abandonment analysis.